Ch 5.1.4-5: More Cross-Validation

Lecture 14 - CMSE 381

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Fri, Oct 6, 2023

Announcements

Last time:

- k-fold CV

This lecture:

- More k-fold CV
- Bias-Variance Tradeoff
- ~~CV for classification~~

Announcements:

- Homework #4 is posted, Due Monday
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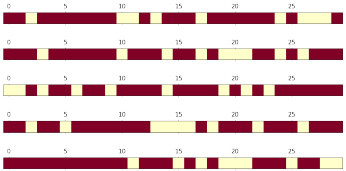
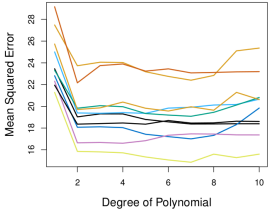
Section 1

k-fold CV

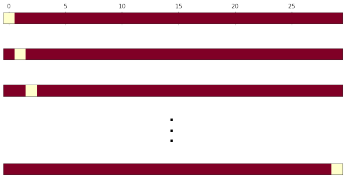
Approximations of Test Error

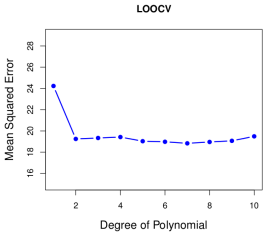
Validation Set





LOOCV

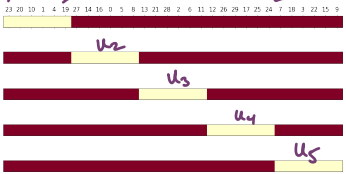


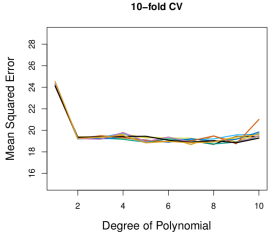


K-fold CV

1st fold U_1

random shuffle



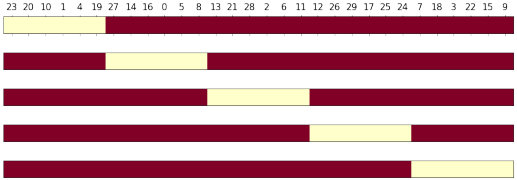


Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

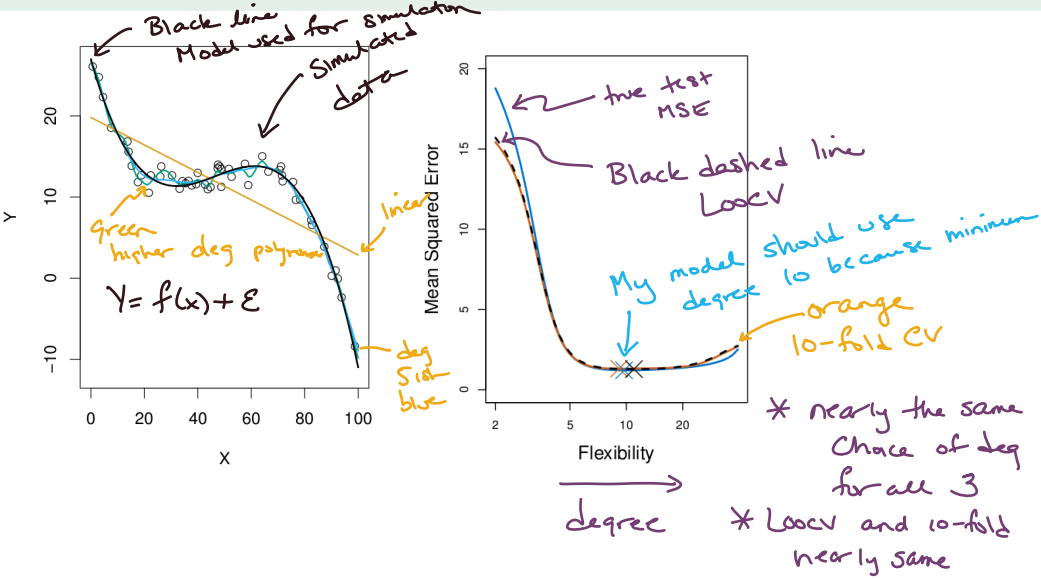
- Rinse and repeat



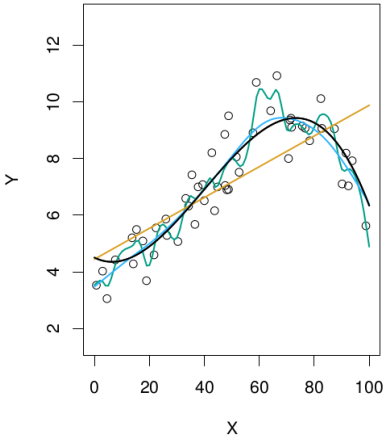
Return

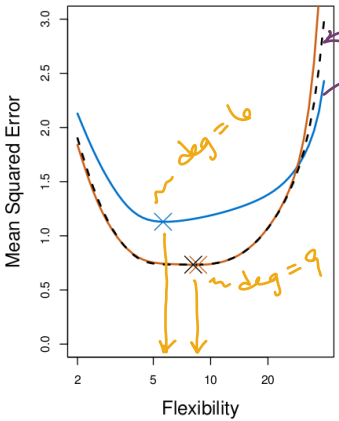
$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

Comparison with simulated data: Ex 3



Comparison with simulated data: Ex 1





orange 10-fold CV

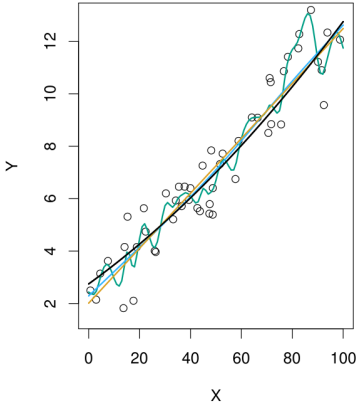
dashed LOOCV

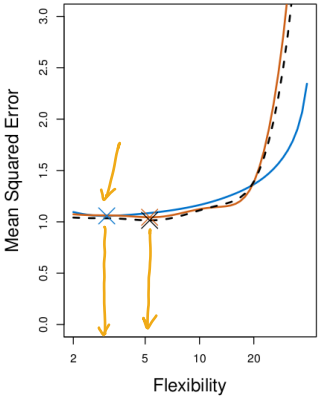
Blue true test error

* True test is diff than LOOCV or 10-fold
* LOOCV or 10-fold generally under estimates of the true test error

I care about $\boxed{\text{where}}$ the minimum occurs not the value it occurs at.

Comparison with simulated data: Ex 2





Takeaways from the examples

* In all plots, LOOCV and 10-fold CV are similar, so reasonable to use 10-fold instead of LOOCV.

* I only care about choose degree (flexability) for my model, so it doesn't matter if the values are off from test error.

Bias-Variance Tradeoff: Bias

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Validation set overestimate test error
b/c only use a small subset of data

k-fold has medium level of bias b/c the
training set has approximately $(k-1)n/k$ observations

LOOCV is better than k-fold just for Bias

LOOCV gives approximately unbiased estimate since all
data pts are used every time

Bias-Variance Tradeoff: Variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

LOOCV has higher variance

* average the outputs of n fitted models but they are all on nearly identical training sets
* high correlation between models

just in variance, k-fold better

k-fold

* k models somewhat less correlated with each other
* mean of many highly correlated models has higher variance than those not correlated

Bias-Variance Tradeoff

~~Added frame from later slide deck, might want to incorporate~~

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Higher Bias

- Validation set overestimates test error b/c used small subset of data
- k-fold gives medium level of bias b/c training set has approximately $(k-1)n/k$ observations
- LOOCV gives approximately unbiased estimate since uses almost all data every time

Lower Bias

Higher Variance

- LOOCV: avg the outputs of n fitted models with high correlation
- k-fold: k fitted models somewhat less correlated with each other

Lower Variance

Usually use $k = 5$ or $k = 10$

Section 2

Aside - Polynomial linear regression

Polynomial regression

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

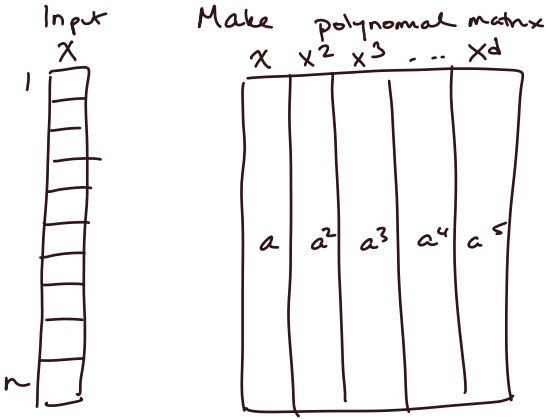
with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \beta_d x_1^d + \varepsilon_i$$

$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_d x_d$

Can learn w/ linear models by passing in x_1^ℓ as predictors

Faking linear regression into doing our work for us



$\xrightarrow{\text{lin reg}}$ get $\beta_0, \beta_1, \ldots, \beta_d$

$$\beta_0 + \beta_1 x^1 + \beta_2 x^2 + \beta_3 x^3 + \ldots$$

Coding - Build a plot for train/test scores vs flexibility

Next time

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	